User Manual for Brain-Controlled Rapid Takeover Device for Drivers Under Autonomous Driving System Failure

1. Product Usage Instructions

This system is based on brain-computer interface technology. It is applicable to scenarios where autonomous driving vehicles move out of the designed operational domain or the system fails due to sudden emergencies (such as road construction, emergency obstacle avoidance). It identifies the driver's takeover intention through single-channel EEG signals (AF3 channel), realizes rapid takeover within 50-100ms, and controls the vehicle to perform operations such as deceleration and left/right lane changing, so as to improve driving safety. The use of the system must strictly follow the core process of "Testing - Formal Application", and the specific operations are as follows:

1.1 Phase 1: Device Connection and Testing (Preliminary Preparation)

Device Connection

In the test environment, connect the single-channel EEG device (TGAM development kit), Arduino Uno microcontroller, Tobii eye tracker, and STM32-NeZha smart car (for simulation testing) through an industrial computer.

The TGAM device is paired with the industrial computer via the HC-05 low-latency Bluetooth module.

The Arduino Uno is connected to the industrial computer via a USB data cable (to realize power supply, program download, and data communication).

Ensure that the data transmission rate of each device is ≥9600bps.

Meanwhile, install the EEG processing software supporting TGAM and the Tobii eye tracker control program.

Function Testing

Run the UC-win/Road driving simulation software to simulate autonomous driving failure scenarios (such as deceleration, left lane changing, right lane changing, emergency obstacle avoidance). Debug the device communication delay and data synchronization accuracy to ensure that the transmission delay of EEG signals, eye movement data, and vehicle motion data (timestamp, 3D position, speed) is ≤100ms and the synchronization error is ≤10ms.

Test the data preprocessing function of Arduino Uno to verify whether it can normally filter EEG data packets and extract raw data, signal strength, attention level, relaxation level, and 8 types of EEG Power band values (Delta, Theta, HighAlpha, etc.).

1.2 Phase 2: Model Training (Executed Only in the Early Stage)

Multimodal Data Collection

Recruit drivers of different driving ages and genders. Let them wear the TGAM device (frontal AF3 channel, ensuring that the electrode sheet is in close contact with the skin), operate the test equipment in the simulated failure scenarios, and collect the drivers' EEG signals.

At the same time, start the Tobii eye tracker to record the 3D coordinates of the fixation point and eye movement types (saccade, pursuit, stillness), and automatically eliminate the takeover data segments interfered by blinking and head shaking.

Collect vehicle motion data (accelerator/brake pedal input value, steering wheel angle) through the simulation software. Based on world time, synchronize the EEG, eye movement, and vehicle data with timestamps to form a training dataset.

Model Construction and Optimization

Based on the recursive feature elimination algorithm, select 14 optimal indicators (including raw data, Delta waves, HighAlpha waves, average power spectral density, etc.) from the 30 collected EEG features as the input features of the model.

Construct a takeover intention recognition model based on the LSTM neural network. Set the ratio of the training set to the test set as 8:2, and use the Adam optimization algorithm to accelerate convergence. Train the model to recognize four types of intentions: "deceleration, left lane changing, right lane changing, invalid command" to ensure high model accuracy. After completing the training, solidify the model parameters, and use different parameters for different types of roads.

1.3 Phase 3: Formal Application (Deployment on Autonomous Driving Vehicles)

Preparation Before Application

On the target autonomous driving vehicle, connect only the TGAM EEG device and Arduino Uno microcontroller through the on-board industrial computer. Configure the Arduino serial port parameters, and start the on-board database and data storage module for real-time storage of EEG signals and vehicle motion data.

Preset failure trigger conditions in the on-board system (such as the laser radar detecting a stationary obstacle within 50 meters ahead, the camera identifying road construction that closes the lane) to ensure matching with the model training scenarios.

Real-Time Data Collection and Status Detection

The driver wears the TGAM device. After the system is started, it collects the EEG signals of the AF3 channel in real time. The Arduino Uno preprocesses the signals (to remove interference), extracts the signal strength, attention level, and EEG Power band values, and transmits them to the on-board industrial computer.

The system automatically detects the status of EEG signals: if the signal strength is <150 (full score is 200), the noise ratio is >30%, or the attention level is <50 (standard value 0-100), it is determined as "unsuitable status". It will immediately remind the driver through the on-board display and voice alarm that "EEG signal is abnormal, please take over the vehicle manually", and at the same time pause the brain-controlled process and wait for the driver's manual intervention.

Intention Prediction and Confidence Inspection

If the signal status is qualified, the on-board industrial computer starts the trained LSTM model, reads the real-time EEG signals and vehicle motion data (speed, position) from the database. The model uses 0.004 seconds as the sliding unit and 0.04 seconds as the time window size to predict the driver's takeover intention multiple times, and takes the command with the highest frequency as the preliminary result.

Calculate the average confidence of the prediction result (the average of the maximum prediction probabilities of all sample sequences):

If the confidence is ≥0.7 (patent preset threshold), enter the vehicle control link.

If the confidence is <0.7, trigger an alarm, prompt "low reliability of intention recognition, please take over manually", and at the same time the system automatically starts emergency deceleration to control the vehicle to move closer to the road side.

Vehicle Control and Process Termination

After the confidence is qualified, the on-board industrial computer transmits the command to the actuator of the autonomous driving vehicle via Bluetooth: the deceleration command controls the brake pedal opening, and the left/right lane changing commands synchronously control the steering wheel angle and turn signal, ensuring that the vehicle action response delay is ≤100ms.

After the vehicle completes the takeover action, the system continuously monitors the vehicle status until the vehicle returns to stable driving (such as uniform speed driving, returning to the normal lane), and the brain-controlled takeover process is officially terminated. If the autonomous driving failure is triggered again later, the system repeats the above "data collection - status detection - prediction control" process.

2. Precautions

2.1 Device and Model Specifications

The TGAM device needs to be calibrated before each use. Detect the signal strength of the AF3 channel through the supporting software to ensure it is ≥80, and avoid interference from hair and sweat.

Regularly check the USB connection stability of Arduino Uno to prevent data transmission interruption.

After the model training is completed, if the autonomous driving vehicle or driver group is replaced, it is necessary to re-collect data to fine-tune the model parameters to avoid a decrease in recognition accuracy due to differences in vehicle performance and drivers' EEG characteristics.

2.2 Safety and Emergency Management

During formal application, the driver must hold a valid driver's license and be familiar with the location of the on-board emergency switch. If manual takeover is not performed within 2 seconds after the alarm, the system will automatically trigger emergency parking to reduce the risk of accidents.

During the operation of the system, ensure the continuous power supply of the on-board industrial computer to avoid data loss caused by vehicle flameout. Back up the EEG and vehicle data after daily use for subsequent model optimization.